

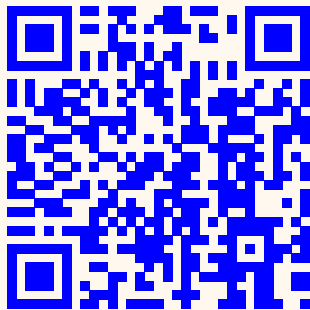
# Torus 1-point functions for the $\mathfrak{sl}(2)$ WZW models and their modular properties

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Joint work with  
Matthew Krauel and Jamal Shafiq  
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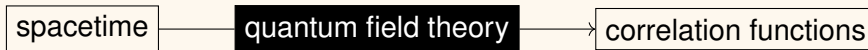
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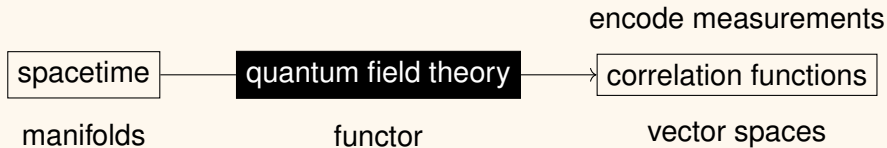
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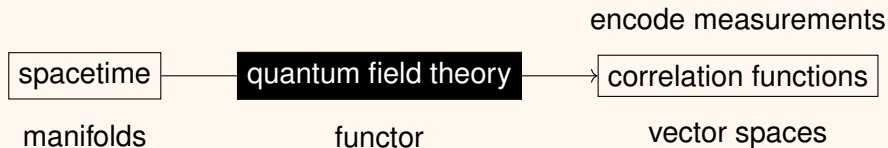
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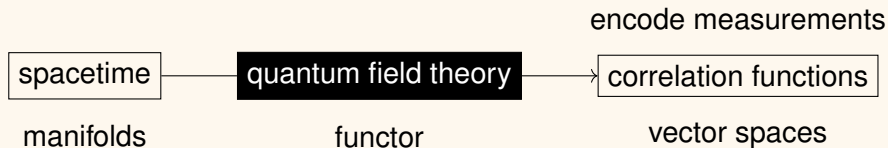
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For 2-dimensional conformally invariant quantum field theories (CFTs) the spacetimes can be

- oriented with Euclidean conformal structure (equivalent to a complex structure),
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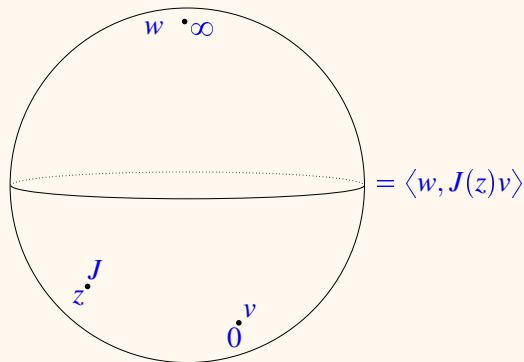
Full local CFTs can be built by glueing together two chiral ones (holomorphic + antiholomorphic = real analytic).

# Motivation: The gist of vertex algebras

A vertex algebra  $V$  is an algebra of “currents”

$$J(z) = \sum_{n \in \mathbb{Z}} J_n z^{-n-1}, \quad J_n \in \text{End}(\mathcal{M}).$$

The variable  $z$  is to be understood as a coordinate on the Riemann sphere  $\mathbb{P}$ .

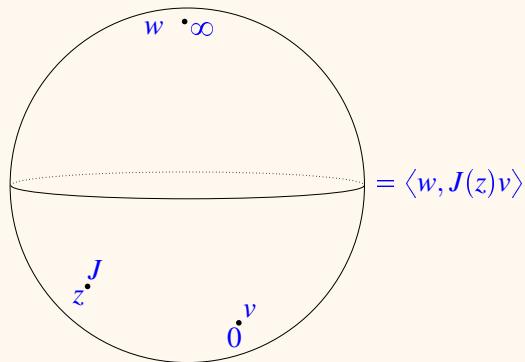


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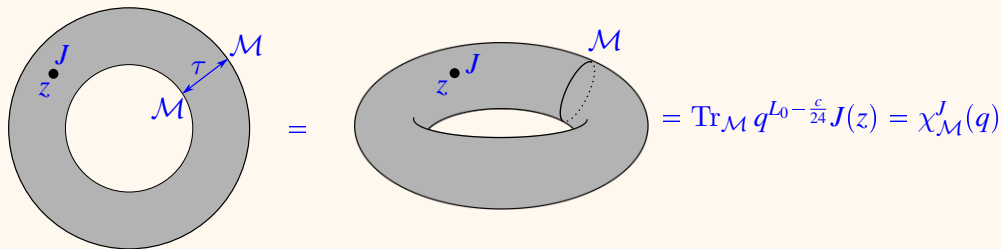
This constructs correlation functions on the Riemann sphere.

But a CFT should be well defined on any Riemann surface!

$$J \in V, v \in \mathcal{M}, w \in \mathcal{M}^*, \mathcal{M} \in V\text{-Mod}.$$

From 3 points on  $\mathbb{P}$  to 1 point on the torus,  $\mathbb{T}$

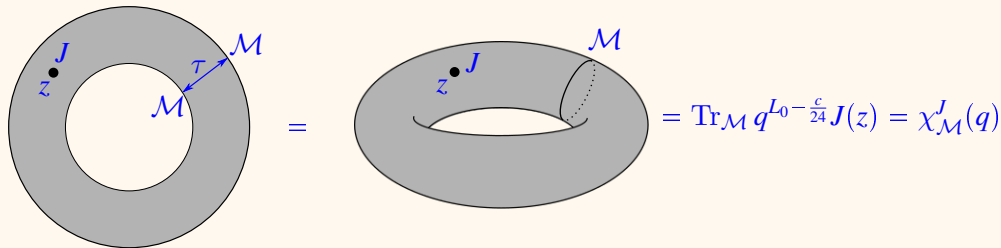
Cut out and glue an annulus in  $\mathbb{P}$ :



Here,  $q = \exp(2\pi i \tau)$  and  $\mathcal{M} \in V\text{-Mod}$ .

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Here,  $q = \exp(2\pi i\tau)$  and  $\mathcal{M} \in V\text{-Mod}$ .

For  $J = \mathbf{1}$  (vacuum), we get the character (graded dimensions) of  $\mathcal{M}$ .

# Complex structures on the torus

Many different  $q = \exp(2\pi i\tau)$  give tori with equivalent complex structures.  
The symmetry groups (mapping class groups) are

empty torus:  $\mathrm{PSL}(2, \mathbb{Z}) = \langle S^2 = (ST)^3, S^2 = 1 \rangle, T(\tau) = \tau + 1, S(\tau) = -\frac{1}{\tau}$

punctured torus:  $B_3 = \langle S^2 = (ST)^3 \rangle$

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- Confusingly, both  $\mathrm{PSL}(2, \mathbb{Z})$  and  $\mathrm{SL}(2, \mathbb{Z})$  are called *the modular group*.
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- For chiral CFT there is a non-trivial action of  $B_3$  on the vector space of torus 1-point functions.
- $B_3$  actions in CFT are projective actions for  $\mathrm{SL}(2, \mathbb{Z})$  and can be “deprojectified” by suitable cocycles.

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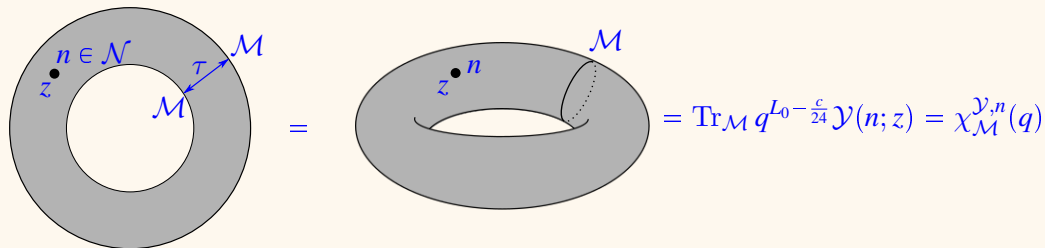
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$$\mathcal{C}^1(V) = \mathrm{span}\{\chi_{\mathcal{M}}^\bullet : \mathcal{M} \in V\text{-Mod}\} \hookrightarrow \mathrm{SL}(2, \mathbb{Z}), B_3$$

# 1-point functions should allow insertions from any module!

Again cut out and glue an annulus in  $\mathbb{P}$ , but with the insertion in a module  $\mathcal{N}$ :



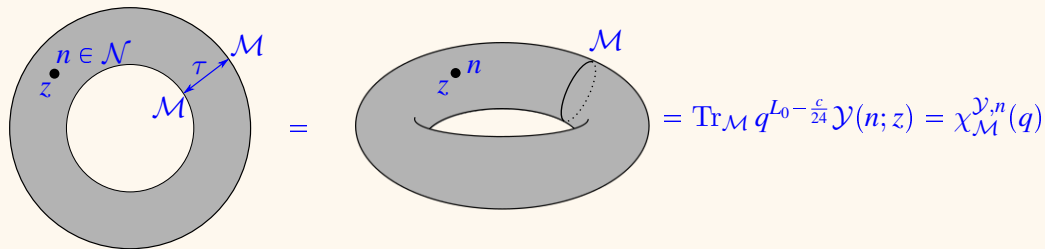
$$= \text{Tr}_{\mathcal{M}} q^{L_0 - \frac{c}{24}} \mathcal{Y}(n; z) = \chi_{\mathcal{M}}^{\mathcal{Y}, n}(q)$$

Here,  $\mathcal{Y}$  is an intertwining operator (a  $V$ -bilinear map).

This can only be non-zero if  $\mathcal{N}$  acts on  $\mathcal{M}$ , i.e. if  $\mathcal{N} \boxtimes \mathcal{M} \twoheadrightarrow \mathcal{M}$ .

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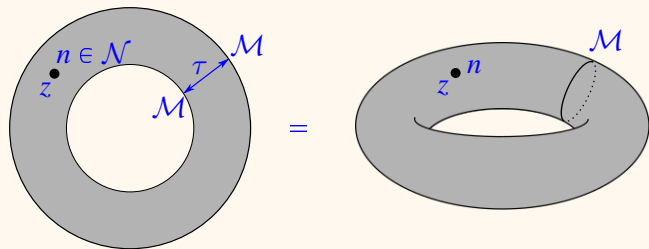
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The diagram illustrates the gluing of an annulus into a torus. On the left, a gray annulus is shown with a central white circle. A point  $z$  is marked on the inner boundary, and a point  $n \in \mathcal{N}$  is marked on the outer boundary. A blue arrow labeled  $\tau$  points from the outer boundary to the inner boundary. The annulus is labeled  $\mathcal{M}$  on both the inner and outer boundaries. This is followed by an equals sign, then a gray torus. A point  $z$  is marked on the torus, and a point  $n$  is marked on the upper boundary. A blue arrow labeled  $\mathcal{M}$  points from the upper boundary to the lower boundary. This is followed by another equals sign, then the expression  $\text{Tr}_{\mathcal{M}} q^{L_0 - \frac{c}{24}} \mathcal{Y}(n; z) = \chi_{\mathcal{M}}^{\mathcal{Y}, n}(q)$ .

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**Challenge:** Can we determine  $\mathcal{C}^1(\mathcal{N})$ ?

# Understanding $\mathcal{C}^1(\mathcal{N})$ from (modular) tensor categories

## Theorem

Explicit formulae for the  $B_3$  generators from categorical data.

$$S^\lambda: \begin{array}{c} \mu \\ | \\ \bullet a \\ / \quad \backslash \\ \lambda \quad \mu \end{array} \mapsto \sum_{\nu, b} S_{\mu, a; \nu, b}^\lambda \cdot \begin{array}{c} \nu \\ | \\ \bullet b \\ / \quad \backslash \\ \lambda \quad \nu \end{array}, \quad S_{\mu, a; \nu, b}^\lambda = \frac{1}{S_{0,0}} \cdot \begin{array}{c} \mu \quad \nu \\ \curvearrowright \quad \curvearrowright \\ a \quad \bar{b} \\ \lambda \end{array}$$

# Understanding $\mathcal{C}^1(\mathcal{N})$ from analytic number theory

Pick a basis  $\{f_1, \dots, f_\ell\}$  of

$$\mathcal{C}_n^1(\mathcal{N}) = \text{span}\{\chi_{\mathcal{M}}^{\mathcal{Y},n}(q) : \chi_{\mathcal{M}}^{\mathcal{Y}}(q) \in \mathcal{C}^1(\mathcal{N})\}$$

then  $\psi_{\mathcal{N}} = (f_1, \dots, f_\ell)^t$  is a vector valued modular form (vvmf) whose weight is the conformal weight of  $n$ :

- $f_i : \mathbb{H} \rightarrow \mathbb{C}$  holomorphic with potential pole at  $i\infty$ .
- For  $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}(2, \mathbb{Z})$

$$f_i|_{\gamma} = \nu(\gamma)^{-1} (c\tau + d)^{-\text{wt } n} f_i \left( \frac{a\tau + b}{c\tau + d} \right),$$

where  $\nu$  is a multiplier system (converts  $\text{B}_3$  actions into  $\text{SL}(2, \mathbb{Z})$  actions).

- $\psi_{\mathcal{N}}|_{\gamma} = \rho(\gamma)\psi_{\mathcal{N}}$ , where  $\rho : \text{SL}(2, \mathbb{Z}) \rightarrow \text{GL}(\ell, \mathbb{C})$  is a representation.

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For  $\mathcal{N} \neq V$ , usually  $\text{wt } n \notin \mathbb{Z}$ , which is highly **non-standard**.

# Vector valued modular forms are well studied in analytic number theory

Common questions:

- Is  $\text{im}(\rho)$  finite? Congruence?
- Let  $M(\rho, k) = \text{span}\{\text{all vvmfs with rep } \rho, \text{ weight } k\}$ . Is  $M(\rho, k)$  finitely generated as a module over the rings of differential modular forms  $R$  or modular functions  $M$ ?
- In the context here: How big is  $\mathcal{C}_n^1(\mathcal{N})$  in  $M(\rho, \text{wt } n)$ ?
- In the context here: Is there information loss when going  $\mathcal{C}^1(\mathcal{N}) \twoheadrightarrow \mathcal{C}_n^1(\mathcal{N})$ ?

## Example: $\mathfrak{sl}(2)$ at level $k \in \mathbb{N}_0$

- Generating currents:  $e, h, f$  with  $[h, e] = 2e, [h, f] = -2f, [e, f] = h$

Operator product relation:  $x(z)y(w) \sim \frac{k\langle x, y \rangle}{(z-w)^2} + \frac{[x, y](w)}{z-w} + \text{reg}$

Normal ordering relation:  $: e(z)^{k+1} := 0$

- Modules: simple highest weight  $\mathfrak{sl}(2)$ -modules

$L(k, \lambda), 0 \leq \lambda \leq k$

- Fusion rules: truncation of Clebsch-Gordan

$$\dim(\lambda \boxtimes \mu \rightarrow \nu) = \begin{cases} 1 & \lambda + \mu + \nu \in 2\mathbb{Z}, |\lambda - \mu| \leq \nu \leq \min\{\lambda + \mu, 2k - \lambda - \mu\} \\ 0 & \text{otherwise} \end{cases}$$

- only even  $\lambda$  can “act” and only on  $\mu$  satisfying  $\frac{\lambda}{2} \leq \mu \leq k - \frac{\lambda}{2}$

- For  $\lambda$  even:  $\dim \mathcal{C}^1(\lambda) = \sum_{\mu=\frac{\lambda}{2}}^{k-\frac{\lambda}{2}} \dim(\lambda \boxtimes \mu \rightarrow \mu) = k - \lambda + 1$

# Specialising from $\mathcal{C}^1(\lambda)$ to $\mathcal{C}_u^1(\lambda)$ , $u \in L(k, \lambda)$

## Question:

Can we find  $u \in L(k, \lambda)$  such that

$$\mathrm{Tr}_{L(k, \mu)} \left[ q^{L_0 - \frac{c}{24}} u_0 \right] = \chi_\mu^u(q) \neq 0?$$

Here,  $u_0$  is the zero mode of the current  $u(z) = \sum_{n \in \mathbb{Z}} u_n z^{-n - \mathrm{wt} u}$ .

- If the  $\mathfrak{sl}(2)$  weight of  $u$  is not 0, then  $\chi_\mu^u(q) = 0$ .
- First “sensible” candidate  $u = f_0^{\frac{\lambda}{2}} |\lambda\rangle$ . However,

$$\left( f_0^{\frac{\lambda}{2}} |\lambda\rangle \right)_0 = \left[ f_0, \left( f_0^{\frac{\lambda}{2} - 1} |\lambda\rangle \right)_0 \right]$$

and traces of commutators vanish because traces are cyclic.

# Non-zero torus 1-point functions

## Lemma

If  $u \in L(k, \lambda)$  generates a simple non-trivial finite  $\mathfrak{sl}(2)$  module, then  $\chi_\lambda^u(q) = 0$ .

## Theorem

- 1 The trivial finite  $\mathfrak{sl}(2)$  module appears for the first time in  $L(k, \lambda)$  at degree  $\frac{\lambda}{2}$ . The multiplicity at this degree is 1.
- 2 The vector spanning this trivial module has the form  $u = f_{-1}^{\frac{\lambda}{2}}|\lambda\rangle + \text{corrections}$ .
- 3  $\chi_\mu^u(q) \neq 0$
- 4 The set  $\{\chi_\mu^u(q) : \frac{\lambda}{2} \leq \mu \leq k - \frac{\lambda}{2}\}$  is linearly independent.
- 5  $\Psi_\lambda(u, q) = (\chi_{\frac{\lambda}{2}}^u(q), \dots, \chi_{k-\frac{\lambda}{2}}^u(q))^t$  is a vvmf.
- 6  $\dim \mathcal{C}_u^1(\lambda) = \dim \mathcal{C}^1(\lambda) = k - \lambda + 1$ , so no information loss from specialising to  $u$ .

$\dim \mathcal{C}^1(\lambda) = 1$ , thus  $\lambda = k$ ,  $k$  even

Let  $\rho_\lambda$  be the representation of  $\mathrm{SL}(2, \mathbb{Z})$  obtained from  $L(k, \lambda)$  and  $\mathcal{V}(\rho_\lambda) = \mathrm{span}\{\Psi_\lambda(w, q) : w \in L(k, \lambda)\}$ .

### Theorem

- 1 The representation  $\rho_k$  is irreducible, finite image and congruence.
- 2  $\rho_k(\mathbf{S}) = \mathbf{e}^{2\pi i \frac{k}{8}}$ ,  $\rho_k(\mathbf{T}) = \mathbf{e}^{2\pi i \frac{k}{24}}$ . Trivial if  $k \in 24\mathbb{Z}$ .
- 3  $\Psi_k(u, q) = \eta^{\frac{3k}{2}}$
- 4  $\mathcal{V}(\rho_k) = R\eta^{\frac{3k}{2}}$
- 5 All vvmfs in  $\mathcal{V}(\rho_k)$  are holomorphic (no poles at  $i\infty$ ).
- 6 As an  $M$  module,  $\mathcal{V}(\rho_k)$  is free and rank 1 with basis  $\{\eta^{\frac{3k}{2}}\}$  for  $k \geq 2$ .
- 7 Graded dimensions of  $\mathcal{V}(\rho_k)$  known with an explicit formula.

$\dim \mathcal{C}^1(\lambda) = 2$ , thus  $\lambda = k - 1$ ,  $k$  odd

Let

$$\Phi = \begin{pmatrix} J^{\frac{1}{24}} {}_2F_1\left(\frac{-1}{24}, \frac{7}{24}; \frac{3}{4}; J^{-1}\right) \\ J^{-\frac{5}{24}} {}_2F_1\left(\frac{5}{24}, \frac{13}{24}; \frac{5}{4}; J^{-1}\right) \end{pmatrix}$$

## Theorem

- 1 The representation  $\rho_{k-1}$  is irreducible, finite image and congruence.
- 2  $\rho_{k-1}(\mathbf{T}) = \text{diag}\left(\mathbf{e}^{2\pi i \frac{k-2}{24}}, \mathbf{e}^{2\pi i \frac{k+4}{24}}\right)$
- 3  $\Psi_{k-1}(u, q) = \eta^{\frac{3k^2+2k-5}{2(k+2)}} \Phi$
- 4  $\mathcal{V}(\rho_{k-1}) = R\Psi_{k-1}(u, q)$ .
- 5 All vvmfs in  $\mathcal{V}(\rho_{k-1})$  are holomorphic for  $k \geq 3$ .
- 6 As an  $M$  module,  $\mathcal{V}(\rho_{k-1})$  is free and rank 2 with basis  $\{\Psi_{k-1}(u, q), \partial\Psi_{k-1}(u, q)\}$  for  $k \geq 2$ .
- 7 Graded dimensions of  $\mathcal{V}(\rho_{k-1})$  known with an explicit formula.

| Level $k$ | Cyclic generator $\Psi_{k-1}(u, q)$                                                                                                                                                                                                                                   |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3         | $q^{3/40} \left( 1 + \frac{1}{5}q - \frac{117}{25}q^2 - \frac{84}{125}q^3 + \frac{3659}{625}q^4 + \cdots \right)$ $q^{13/40} \left( 1 - \frac{9}{5}q - \frac{2}{25}q^2 - \frac{39}{125}q^3 - \frac{126}{625}q^4 + \cdots \right)$                                     |
| 5         | $q^{11/56} \left( 1 - \frac{19}{7}q - \frac{264}{49}q^2 + \frac{6061}{343}q^3 + \frac{22963}{2401}q^4 + \cdots \right)$ $q^{25/56} \left( 1 - \frac{33}{7}q + \frac{247}{49}q^2 + \frac{1672}{343}q^3 - \frac{18183}{2401}q^4 + \cdots \right)$                       |
| 7         | $q^{23/72} \left( 1 - \frac{17}{3}q + \frac{23}{9}q^2 + \frac{3128}{81}q^3 - \frac{13429}{243}q^4 + \cdots \right)$ $q^{41/72} \left( 1 - \frac{23}{3}q + \frac{170}{9}q^2 - \frac{391}{81}q^3 - \frac{10948}{243}q^4 + \cdots \right)$                               |
| 9         | $q^{39/88} \left( 1 - \frac{95}{11}q + \frac{2340}{121}q^2 + \frac{48165}{1331}q^3 - \frac{2895523}{14641}q^4 + \cdots \right)$ $q^{61/88} \left( 1 - \frac{117}{11}q + \frac{5035}{121}q^2 - \frac{74100}{1331}q^3 - \frac{1011465}{14641}q^4 + \cdots \right)$      |
| 11        | $q^{59/104} \left( 1 - \frac{151}{13}q + \frac{7611}{169}q^2 - \frac{35636}{2197}q^3 - \frac{9959957}{28561}q^4 + \cdots \right)$ $q^{85/104} \left( 1 - \frac{177}{13}q + \frac{12382}{169}q^2 - \frac{383087}{2197}q^3 + \frac{1229442}{28561}q^4 + \cdots \right)$ |
| 13        | $q^{83/120} \left( 1 - \frac{73}{5}q + \frac{1992}{25}q^2 - \frac{18177}{125}q^3 - \frac{224261}{625}q^4 + \cdots \right)$ $q^{113/120} \left( 1 - \frac{83}{5}q + \frac{2847}{25}q^2 - \frac{48472}{125}q^3 + \frac{309009}{625}q^4 + \cdots \right)$                |

$\dim \mathcal{C}^1(\lambda) = 3$ , thus  $\lambda = k - 2$ ,  $k$  even

Let

$$\Phi_k = \begin{pmatrix} J^{\frac{(k+1)}{12(k+2)}} {}_3F_2 \left( -\frac{(k+1)}{12(k+2)}, \frac{11k+14}{24(k+2)}, \frac{19k+30}{24(k+2)}; \frac{3k+7}{4(k+2)}, \frac{1}{2}; J^{-1} \right) \\ J^{-\frac{k+1}{6(k+2)}} {}_3F_2 \left( \frac{k+1}{6(k+2)}, \frac{3k+5}{6(k+2)}, \frac{5k+9}{6(k+2)}; \frac{5k+9}{4(k+2)}, \frac{5}{8}; J^{-1} \right) \\ J^{-\frac{5k+11}{12(k+2)}} {}_3F_2 \left( \frac{5k+11}{12(k+2)}, \frac{9k+19}{12(k+2)}, \frac{13k+27}{12(k+2)}; \frac{3}{2}, \frac{5k+11}{4(k+2)}; J^{-1} \right) \end{pmatrix}.$$

## Theorem

- ①  $\rho_{k-2}$  is irreducible and finite image.
- ② Order of  $\rho_{k-2}(\mathbb{T}) = 12(k+2)$  if  $k \in 4 + 6\mathbb{Z}$  or  $4(k+2)$  otherwise.  $\rho_{k-2}$  is non-congruence if  $o(\rho_{k-2}(\mathbb{T})) \nmid 25,401,600 = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7^2$ .
- ③  $\Psi_{k-2}(u, q) = \eta^{\frac{3k^2-2k-8}{2(k+2)}} \Phi_k$  and  $\mathcal{V}(\rho_{k-2}) = R\Psi_{k-2}(u, q)$ .
- ④ All vvmfs in  $\mathcal{V}(\rho_{k-2})$  are holomorphic for  $k \geq 4$ .
- ⑤ As an  $M$  module,  $\mathcal{V}(\rho_{k-2})$  is free and rank 3 with basis  $\{\Psi_{k-2}(u, q), \partial\Psi_{k-2}(u, q), \partial^2\Psi_{k-2}(u, q)\}$  for  $k \geq 4$ .
- ⑥ Graded dimensions of  $\mathcal{V}(\rho_{k-2})$  known with an explicit formula.

| Level $k$ | $\Psi_{k-2}(u, q)$                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4         | $q^{1/24} \left( 1 - \frac{6991}{171}q - \frac{1462930981}{198531}q^2 - \frac{11520966474250}{5360337}q^3 - \frac{467661528323716250}{627159429}q^4 + \dots \right)$ $q^{1/4} \left( 1 + \frac{134}{9}q + \frac{167509}{81}q^2 + \frac{24672291010}{45927}q^3 + \frac{2054193740460070}{11986947}q^4 + \dots \right)$ $q^{13/24} \left( 1 - \frac{31}{27}q + \frac{473}{1215}q^2 - \frac{27056}{32805}q^3 - \frac{1533931}{2657205}q^4 + \dots \right)$                                  |
| 6         | $q^{5/32} \left( 1 - \frac{1041}{20}q - \frac{28822341}{3040}q^2 - \frac{34699584029}{12160}q^3 - \frac{2170275413391777}{2140160}q^4 + \dots \right)$ $q^{3/8} \left( 1 + \frac{74}{5}q + \frac{317943}{130}q^2 + \frac{8423595}{13}q^3 + \frac{21692516271}{104}q^4 + \dots \right)$ $q^{21/32} \left( 1 - \frac{31}{8}q + \frac{423}{128}q^2 + \frac{14247}{7168}q^3 - \frac{485683}{229376}q^4 + \dots \right)$                                                                      |
| 8         | $q^{11/40} \left( 1 - \frac{46803}{775}q - \frac{14944931541}{1375625}q^2 - \frac{574656427747084}{171953125}q^3 - \frac{782261040133149248781}{649123046875}q^4 + \dots \right)$ $q^{1/2} \left( 1 + \frac{1704}{125}q + \frac{108483138}{40625}q^2 + \frac{5094872662288}{7109375}q^3 + \frac{5893213005533601}{25390625}q^4 + \dots \right)$ $q^{31/40} \left( 1 - \frac{503}{75}q + \frac{44149}{3125}q^2 - \frac{206842}{78125}q^3 - \frac{420276376}{17578125}q^4 + \dots \right)$ |
| 10        | $q^{19/48} \left( 1 - \frac{44717}{666}q - \frac{14421863479}{1222776}q^2 - \frac{243672512766437}{66029904}q^3 - \frac{68038738170466662661}{50617747584}q^4 + \dots \right)$ $q^{5/8} \left( 1 + \frac{107}{9}q + \frac{2963152}{1053}q^2 + \frac{64959522367}{85293}q^3 + \frac{5516615806491181}{22261473}q^4 + \dots \right)$ $q^{43/48} \left( 1 - \frac{259}{27}q + \frac{8110}{243}q^2 - \frac{251140}{6561}q^3 - \frac{25036652}{531441}q^4 + \dots \right)$                    |

$$\dim \mathcal{C}^1(\lambda) \geq 4$$

### Theorem

Let the level be  $k = p^t - 2$ , where  $p > 3$  is prime and  $t$  is a positive integer. For  $2 \leq \lambda \leq k$  with  $\lambda$  even, if the representation  $\rho_\lambda$  is irreducible, then it is non-congruence if  $t = 1$  or if  $t > 1$  and  $\lambda + 1 > p^{t-2}$ .

### Proposition (using the categorical formulae)

Choose  $p = 7$ ,  $t = 1$  above and hence  $k = 5$ . Then for  $\lambda = 2$ , the representation  $\rho_2$  is 4-dimensional, irreducible and non-congruence.

### Theorem

Let  $\lambda = k - 3$ ,  $k$  odd. If  $\mathcal{H}(\rho_{k-3}, \nu_{\frac{-k^2+4k+6}{2(k+2)}})$  (the space of holomorphic vvmfs) is a cyclic  $R$ -module, then

$$\mathcal{V}^u(\rho_\lambda) = \mathcal{V}(\rho_\lambda) = \eta^{\frac{3}{2} \frac{k^2-4k-3}{k+2}} \mathcal{H}\left(\rho_{k-3}, \nu_{\frac{-k^2+4k+6}{2(k+2)}}\right).$$

# Outlook

- Analytic number theory tools very helpful up to dimension 3. Some results might generalise up to dimension 5. Bad behaviour expected in higher dimensions.
- Need to find better ways to combine number theory and category theory tools.
- Category theory data well known for  $\mathfrak{sl}(2)$  but higher ranks look hard.
- So far everything has been rational (semisimple and finite). Would be nice to go to finite non-semisimple, but many new tools would be needed.

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Thank you!